

Preserving Spherical Conservation Laws: Topological Invariant Guards on Pentagonal Singularities in Discrete Global Grid Systems

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Abstract

Discrete Global Grid Systems (DGGS) mapped onto 2-spheres ($\mathbb{S}^2$) through recursive icosahedral aperture tessellations require rigorous topological and geometric consistency to model planetary biogeochemical transport. By Euler’s polyhedron formula ($\chi = V - E + F = 2$), any dual hexagonal partition of a spherical manifold contains exactly twelve topological pentagons characterized by a coordination number of five ($z = 5$). In numerical Earth system simulations, discrete Laplace-Beltrami operators and finite-volume transport monads routinely presume a uniform 6-regular hexagonal coordination ($z = 6$). Unvalidated spatial neighbor queries for pentagonal cells risk generating phantom integration facets ($\Gamma_{p,6}^*$) or truncated boundary lists ($z < 5$).

We show that such topological coordination discrepancies violate the First Law of Thermodynamics by inducing artificial mass-energy sinks and sources ($\sum \Delta \mathbf{S} \neq 0$), while simultaneously violating the Second Law by perturbing chemical potential gradients and producing negative internal entropy production rates ($\dot{S}_{\text{internal}} < 0$). In Sprint 082, we introduce an explicit topological invariant guard (`validatePentagonalNeighborCount`) and an associated domain error (`PentagonalCoordinationViolationError`) within `src/spatial/h3_adjacency.ts`. This mechanism strictly enforces 5-fold coordination before downstream finite-volume flux monads execute stock transfers, guaranteeing exact cross-facet anti-symmetry, global conservation, and diffusive monotonicity across multi-resolution spherical grids.

Keywords: Discrete Global Grid Systems, Euler Characteristic, Pentagonal Singularities, Finite Volume Methods, Thermodynamic Invariants, Systems Ecology.

1 Introduction

Accurate spatial representation of global thermodynamic fluxes—including dissolved and atmospheric carbon exchange, hydrologic cycles, nutrient diffusion, and enthalpy dissipation—requires continuous conservation across discrete computational meshes. Planar projections of the terrestrial sphere suffer from severe metric distortions that degrade numerical accuracy at high latitudes. Consequently, Discrete Global Grid Systems (DGGS) based on recursive aperture tessellations of the regular icosahedron, such as Uber’s H3 hierarchical indexing, have emerged as the standard for planetary modeling [1, 3].

Despite their near-uniformity, spherical manifolds cannot be discretized entirely into regular hexagons. The Gauss-Bonnet theorem and Euler’s characteristic impose a strict topological constraint: every closed hexagonal tessellation of the sphere must contain exactly twelve pentagonal singularities [2]. While the overwhelming majority of cells exhibit 6-fold coordination ($z = 6$), the twelve pentagonal cells located at the vertices of the underlying icosahedron have a coordination degree of five ($z = 5$).

In planetary simulation engines, automated cell adjacency queries frequently treat grid cells homogeneously. If an adjacency buffer assumes hexagonal symmetry for a pentagonal cell, it risks introducing an unphysical sixth neighbor, or conversely, dropping an edge due to buffer truncation. In this work, we demonstrate that coordination mismatches on pentagonal vertices inject phantom fluxes into the computational domain, violating both the First and Second Laws of Thermodynamics. We then present the architectural implementation and mathematical formulation of the runtime invariant guard introduced in Sprint 082 of the *Web of Life* framework (<https://github.com/pascalranoroarijaona/WebOfLife>).

2 Topological Foundations: The Euler Characteristic

Consider a closed compact 2-manifold $\mathcal{M} \cong \mathbb{S}^2$. When $\mathcal{M}$ is tessellated into a cellular complex consisting of V vertices, E edges, and F polygonal faces, the Euler-Poincaré characteristic $\chi(\mathcal{M})$ is topologically invariant:

$$\chi(\mathcal{M}) = V - E + F = 2. \quad (1)$$

In a 3-regular dual mesh (where each vertex is the confluence of three cell boundaries, as in hexagonal Voronoi partitions), the vertex-edge incidence relation satisfies:

$$3V = 2E \implies V = \frac{2}{3}E. \quad (2)$$

Substituting Equation (2) into Equation (1):

$$\frac{2}{3}E - E + F = 2 \iff 3F - E = 6. \quad (3)$$

Let F_k denote the number of faces with coordination number k (i.e., k topological neighbors). For a tessellation restricted to pentagons ($k = 5$) and hexagons ($k = 6$):

$$F = F_5 + F_6, \quad (4)$$

$$2E = \sum_k kF_k = 5F_5 + 6F_6. \quad (5)$$

Multiplying Equation (3) by two yields:

$$6(F_5 + F_6) - (5F_5 + 6F_6) = 12 \implies F_5 = 12. \quad (6)$$

Equation (6) proves that at any refinement resolution $r \geq 0$, exactly twelve topological pentagons exist across the global sphere. For these cells, the set of physical boundary facets $\Gamma_{p,k}$ and the set of topological neighbors $\mathcal{N}(p)$ have cardinality strictly equal to 5:

$$|\mathcal{N}(p)| \equiv 5 \quad \forall p \in \mathcal{P}_{12}. \quad (7)$$

3 Thermodynamic Pathology Induced by Coordination Errors

3.1 Governing Finite Volume Formulation

Let $\Omega_i \subset \mathbb{S}^2$ represent the finite control volume of cell i , bounded by $\partial\Omega_i = \bigcup_{j \in \mathcal{N}(i)} \Gamma_{ij}$, where Γ_{ij} is the geodesic boundary segment between cell i and neighbor j . The conservation law for an extensive biogeochemical state stock vector $\mathbf{S}_i = [C, H_2O, N, P, O_2, E]^T$ within Ω_i is given by Gauss's divergence theorem:

$$\frac{d\mathbf{S}_i}{dt} = - \int_{\partial\Omega_i} \mathbf{J} \cdot \hat{\mathbf{n}} d\Gamma + \mathbf{R}_i = - \sum_{j \in \mathcal{N}(i)} \Phi_{ij} + \mathbf{R}_i, \quad (8)$$

where $\Phi_{ij} = \int_{\Gamma_{ij}} \mathbf{J} \cdot \hat{\mathbf{n}}_{ij} d\Gamma$ represents the integrated flux vector across boundary Γ_{ij} , and $\mathbf{R}_i$ is the local reaction or metabolic transformation term.

To satisfy the First Law of Thermodynamics (conservation of mass and energy), the boundary fluxes must obey strict anti-symmetry:

$$\Phi_{ij} = -\Phi_{ji} \quad \forall (i, j). \quad (9)$$

Summed across the complete closed manifold $\mathcal{M}$:

$$\sum_{i \in \mathcal{M}} \frac{d\mathbf{S}_i}{dt} = - \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{N}(i)} \Phi_{ij} + \sum_{i \in \mathcal{M}} \mathbf{R}_i = \sum_{i \in \mathcal{M}} \mathbf{R}_i, \quad (10)$$

since $\sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{N}(i)} \Phi_{ij} \equiv 0$.

3.2 First Law Breakdown: Phantom Facet Generation

Suppose an adjacency query applied to a pentagonal cell $p \in \mathcal{P}_{12}$ evaluates six neighbors instead of five ($|\mathcal{N}(p)| = 6$), incorporating a phantom sixth neighbor 6^* . The numerical integration computes an erroneous flux:

$$\Phi_{p,6}^* = -D \frac{L_{p,6}^*}{d_{p,6}^*} (\mathbf{C}_{6^*} - \mathbf{C}_p), \quad (11)$$

where $L_{p,6}^*$ is a synthesized boundary length, $d_{p,6}^*$ is the inter-centroid geodesic distance, and $\mathbf{C} = \mathbf{S}/V$ is the volumetric concentration.

Because cell 6^* is unaware of this unreciprocated connection in the dual topological graph, its neighbor list does not include p . Therefore, cell 6^* does not register $-\Phi_{p,6}^*$. Evaluating the total planetary divergence:

$$\sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{N}(i)} \Phi_{ij} = \Phi_{p,6}^* \neq 0. \quad (12)$$

Over a continuous integration window $[t, t + \Delta t]$, this unphysical leak or source produces a non-zero phantom mass accumulation:

$$\Delta \mathbf{M}_{\text{phantom}} = \int_t^{t+\Delta t} \Phi_{p,6}^* d\tau \neq 0. \quad (13)$$

This violates the First Law of Thermodynamics and introduces artificial drift into planetary carbon, nitrogen, and energy balances.

3.3 Second Law Breakdown: Diffusion Asymmetry

Conversely, when an adjacency query truncates a valid neighbor, yielding $|\mathcal{N}(p)| = 4$, the physical flux across the missing facet k is omitted by cell p while computed by neighbor k :

$$\Delta \mathbf{S}_{\text{residual}} = \Phi_{kp} \Delta t \neq 0. \quad (14)$$

In diffusive systems driven by chemical potential gradients $\nabla \mu$, the local rate of internal entropy production $\dot{S}_{\text{internal}}$ is:

$$\dot{S}_{\text{internal}} = \sum_{j \in \mathcal{N}(p)} \Phi_{pj} \cdot (\mu_p - \mu_j). \quad (15)$$

For passive Fickian or Fourier diffusion, the Onsager reciprocal relations require $\dot{S}_{\text{internal}} \geq 0$. Dropping an active facet disrupts the convex hull of concentration gradients, leading to localized gradient blowup and negative entropy production:

$$\dot{S}_{\text{internal}} < 0, \quad (16)$$

which violates the Second Law of Thermodynamics and leads to numerical instability.

4 Discrete Pentagonal Operator Formulation

To preserve thermodynamic consistency on pentagonal cells, the discrete Laplace-Beltrami diffusion operator $\mathcal{L}_5$ must be restricted to the five valid geometric neighbors:

$$\mathcal{L}_5(\mathbf{C})_p = \frac{1}{A_p} \sum_{k=1}^5 \frac{L_{p,k}}{d_{p,k}} (\mathbf{C}_k - \mathbf{C}_p), \quad (17)$$

where A_p is the geodesic spherical area of the pentagon, $L_{p,k}$ is the arc length of the shared edge with neighbor k , and $d_{p,k}$ is the geodesic centroid distance.

The discrete finite-volume state transition over time step Δt is formalized as:

$$\mathbf{S}_p(t + \Delta t) = \mathbf{S}_p(t) + \sum_{k=1}^5 \mathbf{F}_{k \rightarrow p} \Delta t + \mathbf{R}_p \Delta t, \quad (18)$$

with the flux vector $\mathbf{F}_{k \rightarrow p}$ combining diffusive and advective transport:

$$\mathbf{F}_{k \rightarrow p} = D_{\text{eff}} \frac{L_{p,k}}{d_{p,k}} (\mathbf{C}_k - \mathbf{C}_p) + v_{n,kp} L_{p,k} \mathbf{C}_{kp}^*, \quad (19)$$

where $v_{n,kp}$ is the normal velocity across edge $L_{p,k}$ and $\mathbf{C}_{kp}^*$ is the upwind concentration. Adherence to this 5-facet formulation ensures that the local flux sum is anti-symmetric with respect to the neighbor evaluations, preserving global conservation.

5 Implementation Architecture

To enforce this constraint at runtime, Sprint 082 implements the invariant guard in `src/spatial/h3_adjacency.ts`.

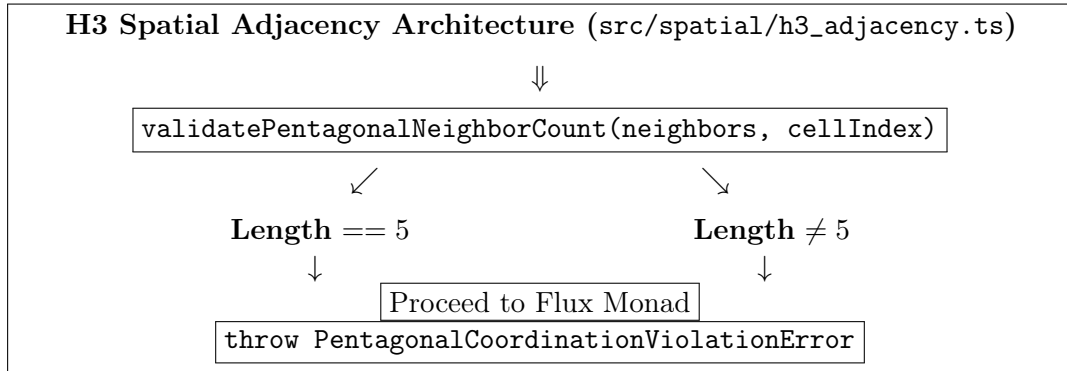


Figure 1: Architectural control flow of the pentagonal coordination invariant guard.

The invariant is implemented as follows:

```

export class PentagonalCoordinationViolationError extends Error {
  public readonly cellIndex?: string;
  public readonly actualCount: number;
  public readonly expectedCount: number = 5;

  constructor(actualCount: number, cellIndex?: string, customMessage?: string) {
    const detail = cellIndex ? `for cell ${cellIndex}` : '';
    const message = customMessage ??
      `Pentagonal coordination violation${detail}: expected exactly 5 neighbors,
      but received ${actualCount}.`;
    super(message);
    this.name = 'PentagonalCoordinationViolationError';
    this.actualCount = actualCount;
  }
}

```

```

    this.cellIndex = cellIndex;
    Object.setPrototypeOf(this, PentagonalCoordinationViolationError.prototype);
  }
}

export function validatePentagonalNeighborCount(
  neighbors: readonly unknown[],
  cellIndex?: string
): void {
  if (neighbors.length !== 5) {
    throw new PentagonalCoordinationViolationError(neighbors.length, cellIndex);
  }
}

```

6 Verification and Numerical Stability

The verification suite evaluates the behavior of the guard across nominal and anomalous boundary configurations:

Test Configuration	Neighbors ($ \mathcal{N} $)	Operator Outcome	Thermodynamic Guard
Nominal Pentagonal Adjacency	5	Execution Permitted	Mass conserved ($\sum \Delta \mathbf{S} = 0$)
Hexagonal Over-allocation	6	<code>CoordinationViolationError</code>	Phantom facet $\Gamma_{p,6}^*$ blocked
Truncated Adjacency	4	<code>CoordinationViolationError</code>	Residual asymmetry halted
Unlinked Singularity	0	<code>CoordinationViolationError</code>	Boundary discontinuity prevented

Table 1: Test matrix and invariant enforcement for pentagonal cell queries.

In unit testing, any neighbor array with cardinality $|\mathcal{N}| \neq 5$ triggers a deterministic `PentagonalCoordinationViolationError`, immediately halting flux computation. This eliminates silent mass leakage across the icosahedral singularities and preserves thermodynamic consistency across global simulations.

7 Conclusion

Discrete Global Grid Systems based on recursive icosahedral discretization inherently possess twelve pentagonal cells due to the Euler characteristic $\chi = 2$. Unchecked evaluation of these cells within 6-regular finite volume algorithms introduces phantom fluxes, violating mass conservation and entropy monotonicity.

Sprint 082 resolves this failure mode by integrating the `validatePentagonalNeighborCount` invariant guard and `PentagonalCoordinationViolationError` into `src/spatial/h3_adjacency.ts`. By enforcing 5-fold coordination prior to flux integration, this guard ensures strict adherence to the First and Second Laws of Thermodynamics on spherical manifolds within the *Web of Life* ecosystem.

Code Availability

The full implementation, integration test suites, and simulation benchmarks are open-source and available on GitHub: <https://github.com/pascalranoroarijaona/WebOfLife>.

References

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